MIL-L-23614B
5 October 1973
SUPERSEDING
MIL-L-23614A
1 February 1967

MILITARY SPECIFICATION

LIGHT, MARKER, DISTRESS, AIRCREW

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 This specification covers requirements for one type and size of a battery powered, waterproofed, aircrew, distress, marker light and the power supply (batteries) for the aircrew, distress, marker light.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein:

SPECIFICATIONS

Federal

V-T-295	Thread, Nylon
PPP-B-566	Box, Folding, Paperboard
PPP-B-636	Boxes, Shipping, Fiberboard
PPP-T-76	Tape, Pressure-Sensitive Adhesive Paper, Water Resistant (For Carton Sealing)

Military

MIL-B-18/224	Battery, Dry, BA-1328/U
MIL-P-116	Preservation, Methods of

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SPECIFICATIONS

Military (Continued)

MIL-T-5422	Testing, Environmental, Aircraft Electronic Equipment
MIL-A-5540	Adhesive, Polychloroprene
MIL-C-19002	Cloth, Coated, and Tape, Coated Cloth-Polychloroprene on Nylon, Pneumatic Life Preserver
MIL-F-21840	Fastener Tape, Hook and Pile, Synthetic

STANDARDS

Federal

FED-STD-151	Metals, Test Methods

Military

MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-129	Marking for Shipment and Storage

DRAWINGS

Bureau of Naval Weapons

63A103D1	Aircrew Distress Marker Light Assembly
63A103D2	Sockets, Contacts, and Wiring Assembly, Aircrew Distress Marker Light
63A103D3	Bayonet Base Lamp and Socket Fixture Details, Aircrew Distress Marker Light
63A103D4	Clip Assembly, Aircrew Distress Marker Light
63A103H5	Case, Aircrew Distress Marker Light
63A103D6	Top and Plug, Aircrew Distress Marker Light
63A103C7	Lens, Aircrew Distress Marker Light

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(When requesting any of the applicable documents, refer to both title and number. Copies of the applicable documents required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 First article - Unless otherwise specified, the aircrew, distress, marker lights furnished under this specification shall be a product which has been inspected and has passed the first article inspection specified in 4.3 through 4.3.2.

3.2 Materials and components - The materials and components shall conform to the applicable specifications, standards, and drawings as listed or required herein. No material or component used in the construction of the light shall have any deleterious effect on itself or on any other material or component used in the construction of the light (see 6.2(g)). Unless otherwise specified, all the elastomeric materials used in the construction of the distress, marker light shall have been manufactured not more than 12 months prior to the date of the delivery of the lights (see 6.2(h)). The lamps shall be new (never used).

3.3 Design - The light shall be a waterproofed unit consisting essentially of a case, one lamp, one lens, and a push-push type switch. The light shall contain a clip and a nylon cord for fastening and securing the light to the clothing or personal equipment of the survivor. The front of the case shall contain hook fastener tape for mating with the pile fastener tape attached to the clothing or personal equipment of the survivor. The light shall also be a unit that is battery powered by two batteries. The arrangement of the components shall conform to 63A103D1.

3.4 Construction - The light shall be constructed in accordance with 63A103D1 through 63A103D4, 63A103H5, 63A103D6, and 63A103C7. Hook fastener tape conforming to MIL-F-21840, Type I, color optional, 1 by 3, plus 0, minus 1/4 inches, shall be superimposed and sewn onto the cloth side of seam tape of the same size conforming to MIL-C-19002, Type III, color optional, as specified in Figure 1. The sides and ends of the hook fastener tape and the seam tape shall be flush with each other. The nylon sewing thread shall conform to V-T-295, Type I or II, Class 1 or 2, Size B, color optional. The coated side of the seam tape shall be cemented to the front of the light as specified in Figure 1 with the adhesive conforming to MIL-A-5540, Class 3.

3.4.1 Power supply - The power supply of each light shall be two batteries conforming to MIL-B-18/224 with the following modification:

(a) Voltage - When inspected as specified in 4.6.1, the voltage at the terminals of each individual battery shall be not less than 1.22 volts.

The light shall not be shipped with the power supply installed nor shall the light and batteries be packaged together.

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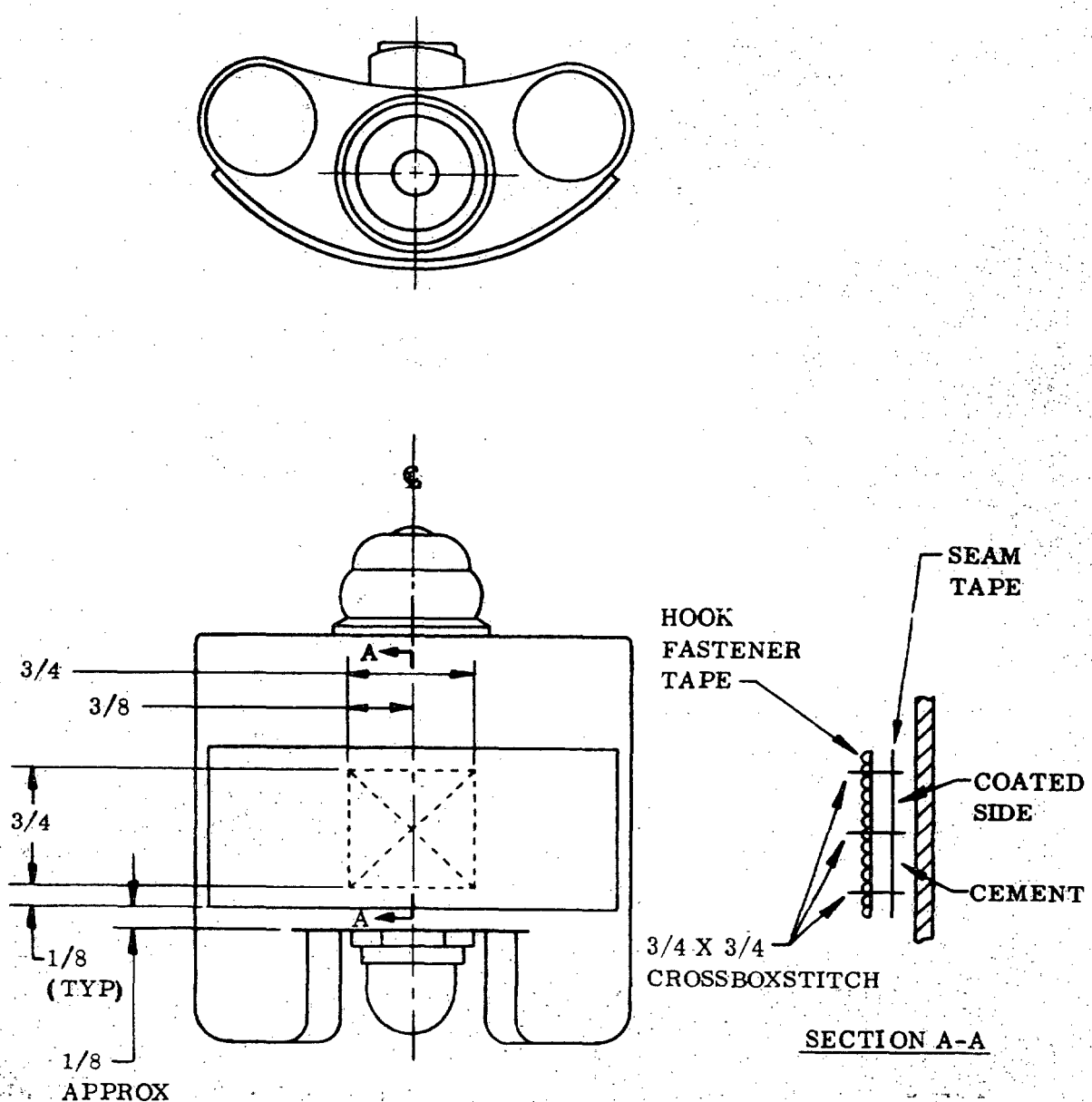


FIGURE 1. ATTACHMENT OF THE HOOK FASTENER TAPE

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3.5 Performance inspections -

3.5.1 Light intensity - When inspected as specified in 4.6.3 through 4.6.3.3, the minimum average light intensity (candlepower) of the horizontal, median, and vertical beams shall be as follows:

Light Beam	Minimum Average Candlepower
Horizontal	1.8
Median	1.9
Vertical	40.0

3.5.2 Expansion due to altitude and waterproofness - When inspected as specified in 4.6.4, the case or any component shall not increase in volume, burst, crack, deform, or leak. The case or any component shall not separate and the lens shall not cloud or fog. All the components shall remain perfectly intact. The light shall then be inspected and conform to the requirements for light intensity, 3.5.1.

3.5.3 Switch activation - When inspected as specified in 4.6.5, the switch shall function properly without any difficulty and the light shall operate.

3.5.4 Exposure cycle - When inspected as specified in 4.6.6, the case or any component shall not burst, crack, or deform. The case or any component shall not separate and the lens shall not cloud or fog. The light shall then be inspected and conform to the requirements for light intensity, 3.5.1, and switch activation, 3.5.3. The exposure cycle inspection shall be performed in sequence with the humidity inspection. The exposure cycle inspection shall be performed first and when completed, the light shall then be inspected for conformance to the requirements for humidity, 3.5.5.

3.5.5 Humidity - When inspected as specified in 4.6.7, the case or any component shall not burst, crack, deform, or separate. The lens shall not cloud or fog. The light shall then be inspected and conform to the requirements for light intensity, 3.5.1, and switch activation, 3.5.3.

3.5.6 Vibration - When inspected as specified in 4.6.8, the light shall not be lighted and the case or any component shall not crack, deform, or separate. The light shall then be inspected and conform to the requirements for light intensity, 3.5.1, and switch activation, 3.5.3.

3.5.7 Salt spray - When inspected as specified in 4.6.9, the light shall not leak or show any evidence of corrosion. The case or any component shall not crack, deform, or separate. The lens shall not cloud or fog. The light shall then be inspected and conform to the requirements for light intensity, 3.5.1, and switch activation, 3.5.3.

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3.5.8 Weight - When weighed as specified in 4.6.10, the assembled light, with the lamp and hook fastener tape installed and without the two batteries, shall weigh not more than 3.25 ounces.

3.6 Color - The color of the various components of the light shall be as specified in the applicable drawings.

3.7 Markings - Each light shall contain the markings and be positioned in the location specified in 63A103D1 and 63A103H5. The markings shall be legible letters and numerals.

3.8 Dimensions - The dimensions of the light shall conform to the applicable drawings and 3.4.

3.9 Workmanship - The distress, marker light shall not be misaligned, deformed, nor contain any rough surface, nick, burr, sharp edge, crack, pit, corrosion, scale, embedded foreign matter, spot, stain, tear, cut, mend, nonspecified hole, burn, detached component, or loose particle. The light shall be uniform in quality and shall be free from mechanical, electrical, or other irregularities or defects which could adversely affect performance, reliability or durability. The lights shall conform to the quality and grade of product established by this specification. The occurrence of defects shall not exceed the acceptance criteria established herein.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection - Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Classification of inspection - The examination and testing of the distress, marker lights shall be classified as follows:

(a) First article inspection - First article inspection consists of examinations and tests performed on samples which are representative of the production item after award of a contract to determine that the production item conforms to the requirements of this specification (see 3.1 and 4.3 through 4.3.2).

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(b) Quality conformance inspection - Quality conformance inspection consists of examinations and tests performed on individual products or lots to determine conformance of the products or lots with the requirements set forth in this specification (see 4.4 through 4.4.1.2).

4.3 First article inspection - The first article inspection of the distress, marker lights shall consist of examinations and tests for all the requirements of this specification.

4.3.1 First article samples - Unless otherwise specified, as soon as practicable after the award of the contract or order, the manufacturer shall submit five lights. The samples shall be representative of the construction, workmanship, components, and materials to be used during production. When a contractor is in continuous production of these lights from contract to contract, submission of further first article inspection samples on the new contract may be waived at the discretion of the procuring activity (see 6.2(c)). Approval of the first article inspection samples or the waiving of the first article inspection does not preclude the requirements for performing the quality conformance inspection. The first article inspection samples shall be furnished to the Government as directed by the contracting officer (see 6.2(d)).

4.3.2 Upon completion of the first article inspection, all the applicable inspection reports and when applicable, recommendations and comments pertinent for use in monitoring production will be forwarded to the cognizant Government activity. One of the approved first article inspection samples of the lights will be returned to the manufacturer, for use in monitoring production. The remaining lights will be destroyed in the first article inspection and shall not be considered as part of the quantity to be delivered under the contract or order.

4.4 Quality conformance inspection - The sampling and inspection levels shall conform to MIL-STD-105. The quality conformance inspection shall consist of the following:

Visual examination of the lights
Dimensional check of the lights
Light intensity
Expansion due to altitude and waterproofness
Switch activation
Weight of the assembled light
Preparation for delivery

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4.4.1 Sampling -

4.4.1.1 Inspection lot -

4.4.1.1.1 Lights - An inspection lot size shall be expressed in units of one light with the lamp installed made essentially under the same conditions and from the same materials and components. The sample unit shall be one light with the lamp installed.

4.4.1.1.2 Preparation for delivery - An inspection lot size shall be expressed in units of one fully prepared shipping container, containing lights with the lamps installed, fully prepared for delivery from essentially the same materials and components. The sample unit shall be one shipping container, containing lights with the lamps installed, fully prepared for delivery with the exception that it need not be sealed.

4.4.1.2 Sampling for examinations and tests of the lights and preparation for delivery - The sample size, acceptance criteria, examinations, and tests required for the lights or preparation for delivery, as applicable, shall be as specified in Table I.

TABLE I

SAMPLE SIZE, ACCEPTANCE CRITERIA, EXAMINATIONS, AND TESTS OF THE LIGHTS AND PREPARATION FOR DELIVERY

INSPECTION	PARAGRAPH		SAMPLE SIZE	ACCEPTANCE CRITERIA
	REQUIREMENT	METHOD		
Visual examination	3.3, 3.4, 3.6, 3.7, 3.9, and applicable drawings	4.6.2.1	Every light for major defects. Inspection Level II for minor defects	Reject all units with any major defect and an acceptable quality level of 6.5 defects per 100 units for minor defects
Dimensional check	3.8	4.6.2.1	Inspection Level S-3	An acceptable quality level of 4.0 defects per 100 units
Light intensity	3.5.1	4.6.3 through 4.6.3.3	Every light	Reject all defective units

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TABLE I (Continued)

INSPECTION	PARAGRAPH		SAMPLE SIZE	ACCEPTANCE CRITERIA
	REQUIREMENT	METHOD		
Expansion due to altitude and water-proofness	3.5.2	4.6.4	Inspection Level S-1	An acceptable quality level of 1.5 defects per 100 units
Switch activation	3.5.3	4.6.5	Every light	Reject all defective units
Weight	3.5.8	4.6.10	Inspection Level S-1	An acceptable quality level of 1.5 defects per 100 units
Preparation for delivery	Section 5	4.6.2.2	Inspection Level S-2	An acceptable quality level of 2.5 defects per 100 units

4.5 Inspection conditions -

4.5.1 Atmospheric conditions - Unless otherwise specified, all the inspections required by this specification shall be conducted at an atmospheric pressure of 28 to 32 inches of mercury and at a temperature of 77 ±18 degrees Fahrenheit (25 ±10 degrees Centigrade). When the inspections are conducted at an atmospheric pressure or temperature different from the above values, proper corrections shall be made for the change in instrument readings.

4.5.2 Synthetic sea water - When sea water is required in an inspection method, synthetic sea water of the composition specified in Table II shall be used. The synthetic sea water shall be prepared by dissolving the sodium sulfate in distilled water and then adding to the solution containing the other dissolved salts. If necessary, an adjustment of the pH to 8.0 shall be made by the addition of sodium hydroxide. The final volume shall be brought up to one liter.

4.5.3 Power supply - When a power supply is required in an inspection method, the power supply conforming to 3.4.1 shall be used.

4.6 Inspection methods -

4.6.1 Voltage - The voltage at the terminals of each battery shall be determined by the use of a voltmeter. The deflection of the voltmeter indicator shall be not less than 25 percent of the full scale. A load of 4.7 ohms shall be applied and the voltage shall be immediately determined (see 3.4.1).

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TABLE II

COMPOSITION OF THE SYNTHETIC SEA WATER

INGREDIENTS	GRAMS PER LITER
Anhydrous Sodium Chloride (NaCl)	22.347
Magnesium Chloride ($MgCl_2.6H_2O$)	9.759
Anhydrous Sodium Sulfate (Na_2SO_4)	3.337
Anhydrous Calcium Chloride ($CaCl_2$)	0.944
Anhydrous Sodium Bicarbonate ($NAHCO_3$)	0.168

4.6.2 Visual examination -

4.6.2.1 Lights - Every light shall be examined visually for major defects to determine conformance to this specification. Each light, selected as a sample unit from the lot, shall be thoroughly checked dimensionally and examined visually for minor defects to determine conformance to this specification. The classification and list of defects, Tables III and IV, as applicable, shall be used to classify and enumerate the defects found.

TABLE III

CLASSIFICATION OF DEFECTS FOR THE VISUAL EXAMINATION OF THE LIGHTS

MAJOR	MINOR
101. Any surface rough, misaligned, or contains any nick, scale, pit, dent, sharp edge, crack, burr, or nonspecified hole.	201. Any marking missing, incorrect, illegible, or incomplete.
102. Any finish missing or any component improperly finished.	202. Color of any component not as specified.
103. Any component missing, malformed, corroded, fractured, or otherwise damaged.	203. Surface unclean or contains any embedded foreign matter.
104. Any component loose, detached, not attached as specified, or otherwise not securely retained. 1/	204. Less than 15 percent of the hooks flattened or missing from the fastener tape.
	205. One or two congealed masses of the adhesive on the surface of the fastener tape greater than 1/4 inch in diameter.

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TABLE III (Continued)

MAJOR	MINOR
105. Any light that does not contain a lamp.	
106. Any power supply installed in the light.	
107. Any functioning part that works with difficulty.	
108. Any spot or stain on the light. 2/	
109. More than 15 percent of the hooks flattened or missing from the fastener tape.	
110. More than two congealed masses of the adhesive on the surface of the fastener tape greater than 1/4 inch in diameter.	
111. Any portion of the seam tape edge not cemented to the case for a distance greater than 1/4 inch in length and 1/8 inch deep.	
112. Any portion of the threads stripped, torn, or broken.	
113. Any threads not as specified.	
114. Lens contains any bubble, striae, check, or scratch. 2/	
115. Any component not as specified or any defect of a component or defect of the assembly not herein classified. 2/	
116. Any component, component part, or required operation omitted, or any operation improperly performed, not herein classified. 2/	

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1/ The snap fastener shall be checked for proper function and attachment by snapping closed and unsnapping the snap fastener at least three times.

2/ The defect shall be classified as a major defect if it affects the serviceability, otherwise it shall be classified as a minor defect.

TABLE IV

LIST OF DEFECTS FOR THE DIMENSIONS OF THE ASSEMBLED LIGHTS

EXAMINE	DEFECT
Measure the individual components and the assembled light without damaging the assembled light.	Any measurement deviating from the dimensions and tolerances as specified in 3.8 shall be enumerated as a dimensional defect.

4.6.2.2 Preparation for delivery - Each of the fully prepared shipping containers, containing lights, selected as a sample unit from the lot, shall be visually examined to determine that the packaging, packing, and marking conform to this specification. The list of defects, Table V, shall be used to enumerate the defects found.

TABLE V

LIST OF DEFECTS FOR THE PREPARATION FOR DELIVERY

ITEM	DEFECT
Exterior and interior markings	Missing, incorrect, incomplete, or illegible; of improper size, location, sequence, or method of application; markings not the same on the interior and exterior containers; paper label missing.
Packaging and packing materials	Any non-conforming component; any component or component part missing, damaged, or otherwise defective.
Workmanship	Inadequate application of the components such as incomplete closure of any container flap or loose strapping; bulging or distortion of any container.
Exterior and interior weight or content	Gross or net weight exceeds the requirement; number per container more or less than specified; lights and power supplies in the same container.

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4.6.3 Light intensity - The equipment for determining the light intensity, candlepower of the horizontal, median, and vertical beams of the light, shall consist of a rotating and tilting apparatus and a calibrated photometer.

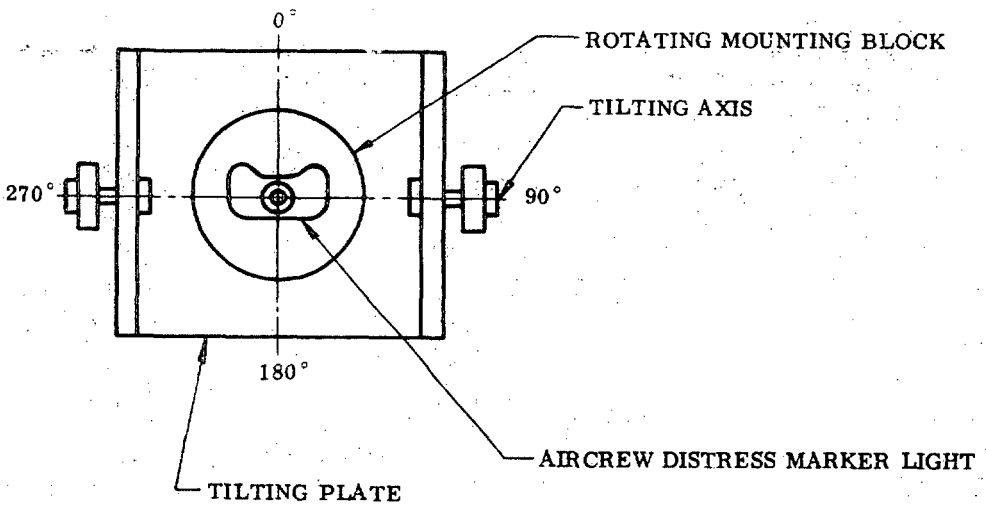
4.6.3.1 Equipment -

4.6.3.1.1 Rotating and tilting apparatus - The apparatus shall consist of a tilting plate and a rotating mounting block. The rotating mounting block shall be designed to securely hold the aircrew, distress, marker light and shall be capable of rotating the distress, marker light around its center axis, perpendicular to the plate surface, independent of the plate orientation. The mounting block shall be secured to the plate in such a manner that the tilting axis of the plate is not more than 1/4 inch above or below the lamp filament. The rotating and tilting apparatus shall conform to Figure 2.

4.6.3.1.2 Photometer - The photometer used shall be calibrated and shall be capable of measuring the candlepower of the various light beams in the ranges specified in 3.5.1. The deflection of the photometer indicator shall be not less than 25 percent of the full scale for any light beam reading. The spectral response of the photometer light sensing element shall approximate the CEI luminous efficiency curve for the spectral response of the human eye. The accuracy of the photometer shall be verified, prior to the start of each inspection lot, by comparison with a standard lamp which has been aged and calibrated in accordance with accepted photometric practice by a recognized established authority.

4.6.3.2 Procedure - The light to be inspected shall contain the power supply (see 4.5.3) and lamp. The rotating and tilting apparatus and the photometer shall be placed on a level, firm, stationary bench or table. The lamp of the aircrew, distress, marker light being inspected and the photometer light sensing element shall be a minimum distance of two feet apart. The lamp filament of the distress, marker light shall be positioned parallel to the plane of the photometer light sensing element. All the light intensity inspections shall be conducted in a dark room environment. Prior to the start of each individual inspection, the photometer indicator shall be checked and adjusted to zero, if necessary. The light shall be securely attached to the rotating and tilting apparatus and activated. The candlepower of each of the horizontal and median annular light beams shall be determined at four positions around the beam, 0, 90, 180, and 270 degrees. The 0 degree is defined as the azimuth position in which an imaginary line, between the photometer light sensing element and the lamp filament of the aircrew, distress, marker light, intersects the long axis of the lamp filament at an angle of 90 degrees. Only one determination for the candlepower of the vertical light beam shall be made. The rotating and tilting apparatus shall be tilted, as required, to superimpose the maximum intensity of the light beam directly upon the photometer light sensing element. The maximum intensity of the light beam reading, at each of the four positions, shall be recorded.

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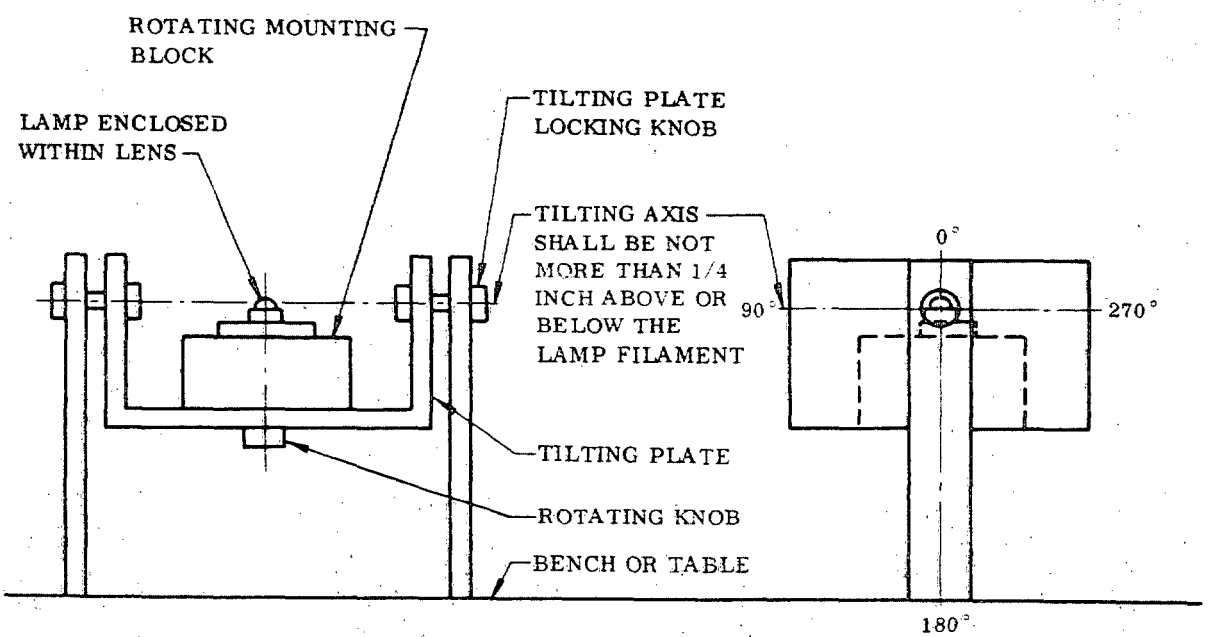


FIGURE 2. LIGHT INTENSITY ROTATING AND TILTING APPARATUS

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4.6.3.3 Report - The candlepower for the horizontal and median light beams shall be the average of the four maximum intensity candlepower readings for each light beam reported to the nearest 0.1 of a candlepower. The candlepower for the vertical beam shall be the individual maximum intensity reading obtained and reported to the nearest 0.1 of a candlepower.

4.6.4 Expansion due to altitude and waterproofness - The light to be inspected shall contain the power supply (see 4.5.3) and lamp. The light shall be placed in a suitable air-tight container. The pressure within the container shall be reduced to an absolute pressure of 75 ±10 millimetres of mercury in a maximum time of 15 minutes. The pressure shall be maintained for 5 ±1 minutes and the light observed for increase in volume, bursting, cracking, deformation, separation of the components, and for fogging or clouding of the lens. Upon completion of the observation, the pressure in the air-tight container shall be returned to normal within one minute and the light examined as aforementioned. Upon completion of the examination and without making any adjustments, the light shall be submerged in the synthetic sea water (see 4.5.2) at a temperature of 77 ±2 degrees Fahrenheit (25 ±1 degrees Centigrade) under a static head of 12 inches of water for 2 ±1/4 hours. The light shall be submerged in a manner that shall permit the water to circulate completely around the light. Upon the expiration of the submersion period, the light shall be removed from the water and the outside surface wiped dry. The light shall then be immediately subjected to the light intensity inspection, 4.6.3 through 4.6.3.3, without removing the lens and lamp or making any adjustment. After the candlepower for each of the light beams has been determined, the lens and each battery plug and battery shall be removed. The lens and the inside of the light shall then be examined for moisture, separation of the components, and bursting, cracking, deformation, and increase in volume of any component. Any evidence of moisture within the lens or inside the light is an indication that the light leaks and is not waterproof.

4.6.5 Switch activation - The light shall contain the power supply (see 4.5.3) and lamp. The switch shall be depressed and released. The light shall have been activated. The switch shall be depressed again and released. The light shall go out. The aforementioned shall be repeated for a total of 10 times. Upon completion of the switch activation for 10 times, the switch shall be examined to determine compliance with 3.5.3.

4.6.6 Exposure cycle - The light to be inspected shall contain the lamp but not the power supply. The light shall be subjected, in the sequence listed, to the following conditions:

2 hours at minus 30 ±5 degrees Fahrenheit (minus 34 ±2.8 degrees Centigrade).
2 hours at 75 ±5 degrees Fahrenheit (24 ±2.8 degrees Centigrade).

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2 hours at 160 ±5 degrees Fahrenheit (71 ±2.8 degrees Centigrade).
12 to 18 hours at 75 ±5 degrees Fahrenheit.

These conditions shall be repeated for a total of five cycles. The amount of elapsed time between each condition and cycle shall be not more than 30 minutes. The light shall be exposed in such manner that the conditioned air circulates completely around the light. Upon completion of the temperature cycles, the power supply (see 4.5.3) shall be installed and, without removing the lens and lamp or making any adjustment, the light shall be subjected to the light intensity inspection, 4.6.3 through 4.6.3.3. After the candlepower for each of the light beams has been determined, the lens and each battery plug and battery shall be removed. The light shall then be examined for cracks, breaks, deformation, separation of the components, and for fogging or clouding of the lens. Upon completion of the examination, the light shall be reassembled and subjected to the switch activation inspection, 4.6.5. Upon completion of the testing and examination, the light, containing the same power supply and lamp, shall be subjected to the humidity inspection, 4.6.7. The lights represented by the inspection sample shall be rejected if the inspection sample does not comply with 3.5.4.

4.6.7 Humidity - The light, containing the power supply and lamp, which was previously subjected to the exposure cycle inspection, 4.6.6, shall be exposed to humidity as specified in MIL-T-5422, except that the preconditioning is not applicable. The light shall be exposed in such a manner that the conditioned air and humidity circulates completely around the light. Upon completion of the humidity exposure, the outer surface of the light shall be wiped dry and immediately subjected to the light intensity inspection, 4.6.3 through 4.6.3.3, without removing the power supply, lens, and lamp or making any adjustment. After the candlepower for each of the light beams has been determined, the lens and each battery plug and battery shall be removed. The light shall then be examined for cracks, bursting, deformation, separation of the components, and for fogging or clouding of the lens. Upon completion of the examination, the light shall be reassembled and subjected to the switch activation inspection, 4.6.5. The lights represented by the inspection sample shall be rejected if the inspection sample does not comply with 3.5.5.

4.6.8 Vibration - The light to be inspected shall contain the power supply (see 4.5.3) and lamp. The light shall be mounted on a vibration inspection stand and vibrated at a double amplitude of 0.03 ±0.006 inch. The frequency shall be varied uniformly from 10 to 55 cycles per second and returned to 10 cycles per second in approximately one minute cycles. The vibration shall be applied for a period of 1 hour in each of three mutually perpendicular planes. The light shall not have been activated (lighted) while being vibrated. Upon completion of the vibration and without making any adjustment, the light shall be subjected to the light intensity inspection, 4.6.3 through 4.6.3.3, without removing the power supply, lens, and lamp. After the candlepower for each of the light beams has been determined, the lens and each battery plug and battery shall be removed. The light shall then be

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examined for cracks, breaks, deformation, and separation of the components. Upon completion of the examination, the light shall be reassembled and subjected to the switch activation inspection, 4.6.5. The lights represented by the inspection sample shall be rejected if the inspection sample does not comply with 3.5.6.

4.6.9 Salt spray - The light to be inspected shall contain the power supply (see 4.5.3) and lamp. The light shall be exposed to 96 hours of salt spray in accordance with FED-STD-151, Method 811. The light shall be exposed in such a manner that the salt atmosphere circulates completely around the light. At the expiration of the salt spray exposure, the outer surface of the light shall be wiped dry and immediately subjected to the light intensity inspection, 4.6.3 through 4.6.3.3, without removing the power supply, lens, and lamp or making any adjustment. After the candlepower for each of the light beams has been determined, the lens and each battery plug and battery shall be removed. The light shall then be examined for bursting, cracking, deformation, corrosion, separation of the components, and for fogging or clouding of the lens. Upon completion of the examination, the light shall be reassembled and subjected to the switch activation inspection, 4.6.5. The lights represented by the inspection sample shall be rejected if the inspection sample does not comply with 3.5.7.

4.6.10 Weight - The weight of the light with the lamp and hook fastener tape installed and without the two batteries shall be determined on any scale or balance capable of weighing to the nearest 0.01 of an ounce (see 3.5.8).

5. PREPARATION FOR DELIVERY

5.1 Packaging - Packaging shall be Level A or C, as specified (see 6.2(e)).

5.1.1 Level A - The light with the lamp installed and without the power supply shall be preserved and cushioned in accordance with MIL-P-116, Method III. No contact preservative is required. Each light shall be packaged within a 0.018 inch thick Kraft lined chipboard folding box, approximately 2-5/8 inches deep, 1-3/8 inches wide, and 3-1/4 inches long, conforming to PPP-B-566, Variety 1, Style II, Type A, Class a, Notched Tucks. A paper label (see 5.3.1) shall be inserted into each chipboard folding box. Forty of the boxed lights shall be packaged, two deep, five in width, and four in length, within a fiberboard container conforming to PPP-B-636, Type CF or SF, Domestic Class, Variety SW, Grade 125. The container shall be constructed and closed in accordance with the appendix to PPP-B-636. All the seams shall be sealed with 3 inch minimum wide tape conforming to PPP-T-76.

5.1.2 Level C - The light with the lamp installed and without the power supply shall be packaged to afford the minimum degree of protection necessary to prevent deterioration or damage during shipment and normal environmental conditions and commercial modes of transportation.

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5.2 Packing - Packing shall be Level A, B, or C, as specified (see 6.2(e)).

5.2.1 Level A - Two containers (eighty lights), packaged as specified in 5.1.1, shall be packed as specified in 5.2.2, except that the fiberboard container shall be Weather-Resistant Class, Variety SW, Grade V3c or V3s. Each container shall be constructed, closed, and reinforced with flat steel strapping or tape banding in accordance with the appendix to PPP-B-636.

5.2.2 Level B - Two containers (eighty lights), packaged as specified in 5.1.1, shall be packed within a snug fitting fiberboard container conforming to PPP-B-636, Type CF or SF, Domestic Class, Variety SW, Grade 275. The inside dimensions of the container shall be approximately 15-1/2 by 11-9/16 by 7-1/8 inches. Each container shall be constructed and closed in accordance with the appendix to PPP-B-636.

5.2.3 Level C - The packaged lights, which require packing for acceptance by the carrier, shall be packed within exterior type shipping containers in a manner that shall assure safe transportation at the lowest rate to the point of delivery. The shipment shall conform to the minimum requirements of the rules and regulations applicable to the mode of transportation selected.

5.3 Marking - In addition to any special markings required by the contract (see 6.2(f)), the interior and exterior containers shall be marked in accordance with MIL-STD-129 and shall include the date of manufacture (month and year).

5.3.1 Paper label - A lightweight white paper label with legible and durable black characters shall be inserted into each chipboard folding box. The label shall contain the following information:

THIS AIRCREW, DISTRESS, MARKER LIGHT, FSN 6230-892-5192
REQUIRES TWO BATTERIES, FSN 6135-274-4035, TO FUNCTION.

6. NOTES

6.1 Intended use - The aircrew, distress, marker light is intended for use by aircrew personnel to attract the attention of search and rescue parties when in a land or sea emergency situation.

6.2 Ordering data - Procurement documents shall specify the following:

(a) Title, number, and date of this specification.

(b) Quantity desired.

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(c) Whether first article inspection is waived (see 4.3.1).

(d) Name and address of the first article inspection laboratory (see 4.3.1).

(e) Selection of applicable levels of packaging and packing (see 5.1 through 5.1.2 and 5.2 through 5.2.3).

(f) Whether any special markings are required (see 5.3).

(g) Certificate of compliance that no material or component of the light will have a deleterious effect on itself or on any material or component of the light (see 3.2).

(h) Certificate of compliance for the age of the elastomeric materials used in the manufacture of the lights (see 3.2).

6.3 Data - For the information of contractors and contracting officers, any of the data, specified in the applicable documents, listed in Section 2 of this specification, or referenced lower-tier documents, need not be prepared for the Government and shall not be furnished to the Government, unless specified in the contract or order. The data, to be furnished, shall be listed on DD Form 1423 (Contractor Data Requirements List), which shall be attached to and made a part of the contract or order. NavWeps Form 4200/25 (Drawings, Lists, and Specifications Required) shall be attached, where applicable.

Custodians:
Navy - AS
Air Force - 82

Preparing activity:
Navy - AS
(Project No. 6230-0197)

Review activities:
Navy - SH

User activities:
Army - ME
Navy - CG

NOTICE - Review/user information is current as of date of this document. For future coordination of changes to this document, draft circulation should be based on the information in the current Federal Supply Classification Listing of DOD Standardization Documents.

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